Sequential Band Fusion for Hyperspectral Anomaly Detection

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***Abstract*— This article proposes a new approach to hyperspectral band processing for anomaly detection, to be called sequential band fusion (SBF), derived from the band sequential (BSQ) data acquisition format used by a hyperspectral imaging sensor which fuses one single band at a time with previously fused band subset sequentially. In order to realize SBF, four versions, SBF-BSQ, initial band driven SBF (IBD-SBF), band prioritization SBF (BP-SBF), and band selection SBF (BS-SBF), are developed. Furthermore, to validate the sequentially fused results by SBF identical to that produced by combining all joint bands together, its mathematical theory and derivations are also presented. Finally, the full utility of SBF in anomaly detection is demonstrated through extensive experiments.**

***Index Terms*— Anomaly detection, band fusion (BF), band sequential (BSQ), sequential band fusion (SBF).**

I. Introduction

DUE to advances in hyperspectral imaging technology, a hyperspectral image contains enormous data volume and wealthy spectral information that can be used for hyperspectral data exploitation. To best utilize bands, band selection (BS) is generally considered as an effective means of preserving and retaining those bands which carry desired spectral information of interest. However, there are a few issues arising in BS. First of all, it requires the prior knowledge of bands to be selected, n_{BS} and then follows up a BS strategy to select appropriate n_{BS} bands, both of which are generally applications-specified. That is, BS is primarily determined by applications. Thus, a different application may result in a different set of selected bands. In addition, a third issue is that once bands are selected, all other bands will be discarded. In this case, only those selected bands can be used for performance analysis.

Manuscript received January 11, 2021; revised February 23, 2021; accepted March 21, 2021. Date of publication April 8, 2021; date of current version December 16, 2021. This work was supported in part by the National Nature Science Foundation of China under Grant 61971082 and Grant 61890964, and in part by the Fundamental Research Funds for the Central Universities under Grant 3132019341. *(Corresponding author: Fang Li.)*

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Digital Object Identifier 10.1109/TGRS.2021.3069734

This article takes a quite different approach from the traditional BS to address the above issues. It is called sequential band fusion (SBF) derived from a hyperspectral band processing viewpoint as an alternative to BS. Unlike BS, SBF does not select bands, but rather processes bands sequentially band-by-band with no need of knowing n_{BS} and BS strategy. SBF utilizes band sequential (BSQ) acquisition format by a hyperspectral imaging sensor to collect data [1], [2] to perform band processing sequentially. As a result of such a sequential band process, it offers sequential changes in spectral profiles across the wavelength range so that the performance of each band can be evaluated band-by-band for data analysis. Then, appropriate bands can be selected based on their corresponding sequential band profiles.

According to BSQ, users are allowed to process the data, while the data being collected at the same time in a band-by-band manner. In this case, a band process can be implemented in real time as well as sequentially. Several approaches along this line, called either progressive band processing (PBP) [1] or recursive band processing (RBP) [2], have been developed for various applications. For example, PBP of constrained energy minimization (CEM) in [3] for target detection, PBP of anomaly detection [4], linear spectral unmixing [5], RBP of orthogonal subspace projection (OSP) [6], RBP of automatic target generation process (ATGP) [7], and PBP of fast iterative pixel purity index (FIPPI) [8], progressive BS for data compression and transmission [9], progressive spectral dimensionality reduction [10], progressive band expansion and reduction [11], and progressive BS for hyperspectral imaging [12]. All these works actually perform SBF by fusing the currently being acquired band with a band subset consisting of previously processed bands. The band process is carried out one band at a time band-by-band sequentially. Therefore, such PBP and RBP can be referred to as SBF-BSQ.

Despite the fact that the original idea of SBF was developed from the concept of BSQ, the SBF presented in this article is particularly developed to take advantage of BSQ to address BS issues. Most importantly, it enables users to provide sequential profiles of band-fused results to identify specific bands of interest along with their gradual and rapid changes in spectral variations among adjacent bands. These advantages cannot be offered by BS. Four versions are derived as follows to realize SBF.

The first version is called SBF-BSQ which is similar to those developed in [1]–[8]. It starts with the first band as the

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initial band. Then, the fusion process takes place sequentially by fusing the band being acquired with the band subset made up of all previously fused bands.

However, on many occasions using the first band as the initial band may not be a best way to utilize SBF-BSQ. To resolve this issue a second version of BSF is derived, to be called initial band driven SBF (IBD-SBF)-BSQ which includes a band prioritization (BP) criterion to select a best possible initial band, $\mathbf{B}_{\text{inital}}$ to be used as the first band to initialize SBF-BSQ. Since $\mathbf{B}_{\text{inital}}$ is generally unlikely to be the first band, three scenarios of IBD-SBF can be developed to better realize SBF-BSQ. One is forward circular IBD-SBF (FCIBD-SBF) which performs SBF-BSQ starting with $\mathbf{B}_{\text{inital}}$ in a forward manner until it reaches the last band and then returns back to the first band. Finally, it reaches the $\mathbf{B}_{\text{inital}} - 1$ to complete an entire cycle. A second scenario is backward circular IBD-SBF (BCIBD-SBF) which performs SBF-BSQ starting with $\mathbf{B}_{\text{inital}}$ in a backward manner until it reaches the first band and then returns back to the last band. Finally, it reaches the $\mathbf{B}_{\text{inital}} + 1$ to complete an entire cycle. A third scenario is hybrid IBD-SBF (HIBD-SBF) which combines FCIBD-SBF and BCIBD-SBF to fuse bands alternatively. So, the key to implementing IBD-SBF is how to select the $\mathbf{B}_{\text{inital}}$. To take care of this issue, two criteria are particularly designed for this purpose, correlation matrix-based criterion, referred to as R-criterion and OSP-based criterion, referred to as OSP-criterion to implement IBD-SBF.

A third version is to extend IBD-SBF to band prioritization sequential band fusion (BP-SBF) without using BSQ. It makes use of R/OSP-criteria as BP criteria, referred to as correlation matrix-based BP (R-BP)/orthogonal subspace projection-based BP (OSP-BP) to prioritize bands. Finally, SBF is carried out band-by-band according to their BP-ranked priorities. In other words, SBF-BSQ can be further improved by BP-SBF with BP replacing BSQ.

A fourth version is band selection sequential band fusion (BS-SBF) which also uses R/OSP-criteria to select bands to realize SBF. Unlike BP-SBF which requires prioritizing all bands, BS-SBF only selects a number of bands, n_{BS}, for SBF. In this case, the knowledge of n_{BS} must be provided *a priori*.

A main issue resulting from SBF to work effectively is how to make sure that the results of fusing two different bands are identical to that of directly combining these two bands. So, to validate this fusion process to be correct, a detailed mathematical theory for SBF and analysis is further derived.

It should be noted that the key idea of SBF different from data fusion lies on the fact that the fusion of SBF takes place in its processor/operator not on the data to be fused. More specifically, SBF is performed on the processor/operator, which is heavily determined by a particular application, for example, anomaly detection considered in this article. Additionally, as a result of SBF, data collection and fusion can be performed at the same time sequentially without waiting for completing data acquisition, a task that generally cannot be accomplished by data or sensor fusion.

Several novelties derived from this article are summarized as follows.

1) Unlike data fusion, which fuses multisource data at an application level, SBF does not depend on applications directly. But instead, it is determined by the processor or operator to be used to carry out the band processing. This is significantly different from BS requiring the prior knowledge of n_{BS} and utilizing band search strategies.
2) SBF provides sequential band profiles of changes in spectral variations across the wavelength range. It does not require determining n_{BS} but rather considered n_{BS} as a parameter which can be varied and determined by spectral changes. This is different from BS which predetermines, n_{BS} as a constant value.
3) By virtue of SBF users are able to identify bands of interest according to their spectral profiles provided by SBF in terms of spectral variations. More specifically, the greater are spectral variations, the more significant are the bands. Conversely, the less are spectral variations, the higher interband correlation is among the bands; thus, the more likely are the bands to be redundant. This advantage cannot be offered by BS.

II. Band Fusion

In order to explore the utility of SBF in hyperspectral imaging, anomaly detection is selected to demonstrate its application. This is because hyperspectral imaging has shown a great success in anomaly detection via many different approaches [1], [13]–[29], most recently, low-rank matrix and sparse matrix [30]–[32], autoencoder (AE) [33], [34], generative adversarial network (GAN) [35], [36], and game theory [37]. On the other hand, with so much information provided by the data, it may also result in significant redundancy that can be removed without sacrificing or compromising data integrity. To address this issue, a commonly used means is BS which selects those bands sensitive to respond to spectral characteristics of anomalies. Many research works have been devoted to this effort [38]–[42]. SBF offers a new alternative to BS.

Although many anomaly detectors have been proposed such as [13]–[37], most of them are designed based on two classic anomaly detectors. One is called Reed–Xiaoli anomaly detector (RX-AD) developed in [43] and given by

$$\delta^{\text{RX-AD}}(\mathbf{r}) = (\mathbf{r} - \boldsymbol{\mu})^T \mathbf{K}_{L\times L}^{-1}(\mathbf{r} - \boldsymbol{\mu}) \tag{1}$$

where $\mathbf{r} \in \Re^{L\times 1}$ is a data sample currently being processed, μ is the sample mean of the image, and $\mathbf{K}_{L\times L} \in \Re^{L\times L}$ is the global sample data covariance matrix given by $\mathbf{K}_{L\times L} = (1/N)\sum_{i=1}^{N}(\mathbf{r}_i - \boldsymbol{\mu})(\mathbf{r}_i - \boldsymbol{\mu})^T$, N is the total number of pixels, and L is the total number of bands. Since RX-AD in (1) uses the covariance matrix $\mathbf{K}_{L\times L}$, it is called the covariance $\mathbf{K}_{L\times L}$-based AD, referred to as K-AD.

The other is correlation sample matrix $\mathbf{R}$-based anomaly detector developed in [1], [13], and [14], referred to as R-AD

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given by

$$\delta^{\text{R-AD}}(\mathbf{r}) = \mathbf{r}^T \mathbf{R}_{L\times L}^{-1} \mathbf{r} \tag{2}$$

where $\mathbf{r} \in \Re^{L\times 1}$ is a data sample currently being processed, $\mathbf{R}_{L\times L} \in \Re^{L\times L}$ is the global sample correlation matrix given by $\mathbf{R}_{L\times L} = (1/N)\sum_{i=1}^{N} \mathbf{r}_i \mathbf{r}_i^T$. R-AD is called correlation-anomaly detector.

Of particular interest is R-AD which can be easily implemented as a recursive form to be used for SBF [2]. This is because due to the fact that

$$\mathbf{K}_{L\times L} = \mathbf{R}_{L\times L} - \boldsymbol{\mu}\boldsymbol{\mu}^T. \tag{3}$$

K-AD can be actually realized by R-AD. Because of (3), only R-AD is discussed in this article.

According to (1)–(3) the key issue of implementing SBF in anomaly detectors is how to fuse a single band or a band subset used in the correlation matrix $\mathbf{R}_{L\times L}$ in (2). There are two scenarios: 1) SBF of two single bands and 2) SBF of one single band with a band subset and SBF of two band subsets. Nevertheless, it is worth noting that the single band and band subset to be fused are not necessary to be particularly selected and they can be arbitrary.

A. Band Fusion of Two Single Bands

Generally speaking, the fusion of two single bands generally takes place as an initial condition of SBF.

Let $\mathbf{r}_{\mathbf{b}_s} = (r_{1s}, r_{2s}, \ldots, r_{Ns})^T$ and $\mathbf{r}_{\mathbf{b}_t} = (r_{1t}, r_{2t}, \ldots, r_{Nt})^T$ be band vectors of two single-band images $\mathbf{b}_s$ and $\mathbf{b}_t$ at spectral wavelengths s and t, respectively. Also, assume that the data matrix formed by these two band vectors is given by

$$\mathbf{X}_{\mathbf{b}_s\mathbf{b}_t} = \begin{bmatrix} r_{1s} & r_{2s} & \cdots & r_{(N-1)s} & r_{Ns} \\ r_{1t} & r_{2t} & \cdots & r_{(N-1)t} & r_{Nt} \end{bmatrix} \tag{4}$$

$$\begin{aligned} \mathbf{R}_{\mathbf{b}_s\mathbf{b}_t} &= (1/N)\mathbf{X}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{X}_{\mathbf{b}_s\mathbf{b}_t}^T = N^{-1}\begin{bmatrix} \mathbf{r}_{\mathbf{b}_s}^T \\ \mathbf{r}_{\mathbf{b}_t}^T \end{bmatrix}\begin{bmatrix} \mathbf{r}_{\mathbf{b}_s} & \mathbf{r}_{\mathbf{b}_t} \end{bmatrix} \\ &= N^{-1}\begin{bmatrix} \mathbf{r}_{\mathbf{b}_s}^T\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}_{\mathbf{b}_s}^T\mathbf{r}_{\mathbf{b}_t} \\ \mathbf{r}_{\mathbf{b}_t}^T\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}_{\mathbf{b}_t}^T\mathbf{r}_{\mathbf{b}_t} \end{bmatrix} \end{aligned} \tag{5}$$

where

$$\mathbf{R}_{\mathbf{b}_s\mathbf{b}_t}^{-1} = N\begin{bmatrix} \mathbf{r}_{\mathbf{b}_s}^T\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}_{\mathbf{b}_s}^T\mathbf{r}_{\mathbf{b}_s} \\ \mathbf{r}_{\mathbf{b}_t}^T\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}_{\mathbf{b}_t}^T\mathbf{r}_{\mathbf{b}_s} \end{bmatrix}^{-1}. \tag{6}$$

In this case, when $\mathbf{r}$ and $\mathbf{R}$ use only two bands, $\mathbf{b}_s$ and $\mathbf{b}_t$, R-AD fused by $\mathbf{b}_s$ and $\mathbf{b}_t$ can be specified by $\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t}^T\mathbf{R}_{\mathbf{b}_s\mathbf{b}_t}^{-1}\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t}$ in terms of $r_{\mathbf{b}_s}^T R_{\mathbf{b}_s}^{-1} r_{\mathbf{b}_s}$ and $r_{\mathbf{b}_t}^T R_{\mathbf{b}_t}^{-1} r_{\mathbf{b}_t}$. $\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t} = (r_s, r_t)^T$, $R_{\mathbf{b}_s} = (1/N)\sum_{i=1}^{N} r_{is}^2$, and $R_{\mathbf{b}_t} = (1/N)\sum_{i=1}^{N} r_{it}^2$. Appendix A proves that

$$\begin{aligned} \delta_{\mathbf{b}_s\mathbf{b}_t}^{\text{R-AD}} &= \mathbf{r}_{\mathbf{b}_s\mathbf{b}_t}^T R_{\mathbf{b}_s\mathbf{b}_t}^{-1}\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t} \\ &= \delta_{\mathbf{b}_s}^{\text{R-AD}} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\left(\mathbf{r}_{\mathbf{b}_s}^T \upsilon_{\mathbf{b}_s\mathbf{b}_t\mathbf{b}_s} - N\mathbf{r}_{\mathbf{b}_t}\right) \\ &\quad \times\left(\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}^T \mathbf{r}_{\mathbf{b}_s} - N\mathbf{r}_{\mathbf{b}_t}\right) \end{aligned} \tag{7}$$

where $\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} = R_{\mathbf{b}_s}^{-1}\mathbf{X}_{\mathbf{b}_s}\mathbf{x}_{\mathbf{b}_t}$ and $\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} = (\mathbf{x}_{\mathbf{b}_t}^T[P_{\mathbf{X}_{\mathbf{b}_s}^T}^{\perp}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

B. Band Fusion of One Single Band With One Band Subset

The core of SBF is to fuse the current band to be fused with the band subset which consists of all bands being processed. Also, assume that a band set $\mathbf{B}_{s,m} = \{\mathbf{b}_{s_1}, \mathbf{b}_{s_2}, \ldots, \mathbf{b}_{s_m}\}$ and define its associated data matrix by

$$\mathbf{X}_{\mathbf{B}_{s,m}} = \begin{bmatrix} r_{1s_1} & r_{2s_1} & \cdots & r_{(N-1)s_1} & r_{Ns_1} \\ r_{1s_2} & r_{2s_2} & \cdots & r_{(N-1)s_2} & r_{Ns_2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ r_{1s_{m-1}} & r_{2s_{m-1}} & \cdots & r_{(N-1)s_{m-1}} & r_{Ns_{m-1}} \\ r_{1s_m} & r_{2s_m} & \cdots & r_{(N-1)s_m} & r_{Ns_m} \end{bmatrix} \tag{8}$$

$$\mathbf{X}_{\mathbf{B}_{s,m}\mathbf{b}_t} = \begin{bmatrix} r_{1s_1} & r_{2s_1} & \cdots & r_{(N-1)s_1} & r_{Ns_1} \\ r_{1s_2} & r_{2s_2} & \cdots & r_{(N-1)s_2} & r_{Ns_2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ r_{1s_m} & r_{2s_m} & \cdots & r_{(N-1)s_m} & r_{Ns_m} \\ r_{1t} & r_{2t} & \cdots & r_{(N-1)t} & r_{Nt} \end{bmatrix} \tag{9}$$

$$\mathbf{R}_{\mathbf{B}_{s,m}\mathbf{b}_t} = N^{-1}\mathbf{X}_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{X}_{\mathbf{B}_{s,m}\mathbf{b}_t}^T = N^{-1}\sum_{i=1}^{N}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{X}_{\mathbf{B}_{s,m}}^T + \mathbf{b}_t\mathbf{b}_t^T \tag{10}$$

where

$$\mathbf{R}_{\mathbf{B}_{s,m}\mathbf{b}_t}^{-1} = N\left(\sum_{i=1}^{N}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{X}_{\mathbf{B}_{s,m}}^T + \mathbf{b}_t\mathbf{b}_t^T\right)^{-1}. \tag{11}$$

Then, the $(\mathbf{B}_{s,m}, \mathbf{b}_t)$-fused anomaly detector can be described by $\mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t}^T\mathbf{R}_{\mathbf{B}_{s,m}\mathbf{b}_t}^{-1}\mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t}$ in terms of $\mathbf{r}_{\mathbf{B}_{s,m}}^T\mathbf{R}_{\mathbf{B}_{s,m}}^{-1}\mathbf{r}_{\mathbf{B}_{s,m}}$ and $r_t^T R_t^{-1} r_t$ where $\mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t} = (\mathbf{r}_{\mathbf{B}_{s,m}}^T, \mathbf{r}_{\mathbf{b}_t})^T$ and $R_t = (1/N)\sum_{i=1}^{N} r_{it}^2$. Appendix B proves that

$$\begin{aligned} \delta_{\mathbf{B}_{s,m}\mathbf{b}_t}^{\text{R-AD}} &= \delta_{B_{s,m}}^{\text{R-AD}} \\ &\quad + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\left(\mathbf{r}_{\mathbf{B}_{s,m}}^T \upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} - N\mathbf{r}_{\mathbf{b}_t}\right) \\ &\quad \times\left(\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}^T \mathbf{r}_{\mathbf{B}_{s,m}} - N\mathbf{r}_{\mathbf{b}_t}\right) \end{aligned} \tag{12}$$

where $\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} = \mathbf{R}_{\mathbf{B}_{s,m}}^{-1}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{x}_{\mathbf{b}_t}$, and $\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} = (\mathbf{x}_{\mathbf{b}_t}^T[P_{\mathbf{X}_{\mathbf{B}_{s,m}}^T}^{\perp}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

III. Sequential Band Fusion

Apparently, from the derivations of SBF given in Section II, the single band and band subsets used for SBF can be arbitrary. As a result, one primary advantage of SBF is that these band subsets can be determined or selected at users' discretion. Specifically, the techniques developed in Sections II-A and II-B can be implemented for SBF to derive the following four versions for SBF.

A. Sequential BF-BSQ

The first approach developed for band fusion (BF) was previously studied in [1]–[8], referred to as PBP, which is derived from the BSQ format used by a hyperspectral imaging sensor to collect data band-by-band [1]. It starts with the first band as an initial band and then adds one band at a time sequentially until it reaches the last band. As a matter of fact, SBF-BSQ can be considered as a special case of SBF presented in Section II-B, which fuses one single band currently

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being acquired by BSQ with a band subset which comprises previously added bands. Two advantages can be gained from SBF. One is that SBF can provide sequential band profiles of spectral changes obtained by BSQ-acquired bands across spectral wavelengths. The other is that SBF does not require any BS technique and thus, there is no need to determine the number of bands to be selected, nBS. Several applications have been also investigated for using PBP in [1]–[12].

B. IBD SBF

As noted above, SBF-BSQ is implemented according to BSQ which generally starts the first band as its initial band. An advantage of doing so is that it can be implemented in real time. But this is traded for a disadvantage that the first band is generally not the best band to start with. A judicious selection of an initial band determines how effective the SBF performs in sequence. It is similar to an iterative algorithm whose performance is also determined by its initial condition. If the first band is selected appropriately, SBF will achieve its performance with fewer fused bands than SBF-BSQ using the first band as the initial band. SBF using a particularly selected initial band as its first band is called IBD-SBF. The price to be paid by IBD-SBF is that it cannot be implemented in real time as SBF-BSQ does.

Since the selected initial band may not be the first band, there are three scenarios to perform SBF.

1) Forward Circular IBD-SBF: One is FCIBD-SBF which performs SBF-BSQ in a forward and circular manner to run through all bands. That is, assume the initial band is selected as band number l^* and the last band number is L. FCIBD-SBF starts $\mathbf{b}_{l*}$ as its first band and then proceeds its next band, $\mathbf{b}_{l*+1}$ until it reaches the last band $\mathbf{b}_L$. It then continues on in a circular manner by returning its first band until it reaches $\mathbf{b}_{l*-1}$, i.e., $\mathbf{b}_{l*}, \mathbf{b}_{l*+1}, \ldots, \mathbf{b}_L, \mathbf{b}_1, \ldots, \mathbf{b}_{l*-1}$ and finally terminates its process.

2) Backward IBD-SBF: In contrast to FIBD-SBF, a second one is called BCIBD-SBF which performs SBF-BSQ in a backward and circular manner. Like FCIBD-SBF BIBD-SBF can be also implemented circularly to run through all bands, $\mathbf{b}_{l*}, \mathbf{b}_{l*-1}, \ldots, \mathbf{b}_2, \mathbf{b}_1, \mathbf{b}_L, \ldots, \mathbf{b}_{l*+1}$. That is, it starts with $\mathbf{b}_{l*}$. When it reaches the first band and then returns to the last band $\mathbf{b}_L$ and continues on until it reaches $\mathbf{b}_{l*+1}$ and terminates its process.

3) Hybrid IBD-SBF: A third one is called HIBD-SBF which proceeds bands forward and backward alternatingly until it reaches either the last band L, $\mathbf{b}_{l*+1}, \mathbf{b}_{l*-1}, \mathbf{b}_{l*+2}, \mathbf{b}_{l*-2}, \ldots, \mathbf{b}_1, \ldots, \mathbf{b}_{L-1}, \mathbf{b}_L$ if $l^* \leq L/2$ or the first band, $\mathbf{b}_{l*+1}, \mathbf{b}_{l*-1}, \mathbf{b}_{l*+2}, \mathbf{b}_{l*-2}, \ldots, \mathbf{b}_L, \ldots, \mathbf{b}_2, \mathbf{b}_1$ if $l^* > L/2$.

There are two significant differences between IBD-SBF and SBF-BSQ. One is that IBD-SBF requires receiving all bands and finding the best initial band prior to its sequential band process. In this case, IBD-SBF cannot be implemented in real time as SBF does. Nevertheless, once the initial band is determined, then IBD-SBF can be still implemented as SBF-BSQ. The other is that IBD-SBF can be also considered as SBF using a circular BSQ process.

C. BP-SBF

To resolve the initial band issue, a third approach is presented as an alternative to IBD-SBF, to be called BP-SBF, which extends SBF by including BP to prioritize all the bands prior to SBF. The first work was explored in progressive BS for satellite hyperspectral data compression and transmission in [9], progressive dimensionality reduction by projection index-based projection pursuit (PDR-PIPP) in [10] and [45, Chapter 21], progressive band dimensionality process (PBDP) by BP criteria for endmember finding [11] and progressive band selection (PBS) for spectral unmixing [12], all of which perform BP-SBF using BP as a criterion to rank spectral bands so that the data can be processed band-by-band one band at a time progressively. Two types of BP are particularly designed for BP-SBF, R-BP, and OSP-BP.

1) Correlation Matrix-Based BP: The R-BP presented here is derived from constrained BS (CBS) developed in [45] with R specified by the global correlation matrix $\mathbf{R}$. By taking advantage of CBS, we can custom design a new criterion from (2) as anomaly detection-based BP defined by

$$\rho_l^{\mathbf{R}} = \mathbf{b}_l^T \mathbf{R}_{\mathbf{\Omega}-\mathbf{b}_l}^{-1} \mathbf{b}_l = \left\| \mathbf{R}_{\mathbf{\Omega}-\mathbf{b}_l}^{-1/2} \mathbf{b}_l \right\|^2 \tag{13}$$

to prioritize all the spectral bands $\mathbf{\Omega} = \{\mathbf{b}_l\}_{l=1}^L$. That is, the greater value is $\rho_l^{\text{R-AD}}$, the higher priority is the lth band $\mathbf{b}_l$. As a result of $\rho_l^{\text{R-AD}}$ all bands can be prioritized according to their decreasing values.

2) OSP-Based BP: Another criterion that can be used for BP-SBF is derived from OSP [46], [47]. Suppose that $\mathbf{U}$ is defined by $\mathbf{U} = \mathbf{\Omega} - \mathbf{b}_l$. Then we can define a $\mathbf{U}$-annihilator $P_{\mathbf{U}}^{\perp}$ by

$$P_{\mathbf{U}}^{\perp} = \mathbf{I} - \mathbf{U}\mathbf{U}^{\#} = \mathbf{I} - \mathbf{U}\left(\mathbf{U}^T\mathbf{U}\right)^{-1}\mathbf{U}^T. \tag{14}$$

By virtue of (14) OSP-based criterion can be further defined and derived as follows:

$$\eta_l^{\text{OSP}} = \mathbf{b}_l^T P_{\mathbf{U}}^{\perp} \mathbf{b}_l = \left\| \mathbf{b}_l^T P_{\mathbf{U}}^{\perp} \right\|^2 \tag{15}$$

which is similar to (13).

3) How to Select Initial Band as First Band by SBF: As noted in IBD-SBF the selection of an initial band, $\mathbf{b}_{l*}$ is crucial. According to (13) and (15), we can further implement SBF with its initial bands selected as the first band to be processed by SBF based on

$$\mathbf{b}_{\text{initial}}^{\mathbf{R}} = \arg\left\{ \max_{1 \leq l \leq L} \rho_l^{\text{R-AD}} \right\} \tag{16}$$

$$\mathbf{b}_{\text{initial}}^{\text{OSP}} = \arg\left\{ \max_{1 \leq l \leq L} \eta_l^{\text{OSP}} \right\}. \tag{17}$$

D. BS-SBF

A fourth approach is BS-SBF which utilizes a band search strategy to select bands. In particular, BS-SBF does not prioritize bands as BP-SBF does. Instead, it replaces BP with a band search strategy to select bands to be used for BSF. Consequently, it requires the prior knowledge of the number of bands needed to be selected, n_{BS}. Nevertheless, to implement BS-SBF, any existing BS method can be used in conjunction with SBF. Specifically, two BP criteria in (13) and (15) can be used as BS search strategies to derive BS-SBF as follows, both of which have not been investigated yet in the literature.

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BS-SBF

1. Initial condition:

For a given n_{BS}, select the initial band $\mathbf{b}_{\mathrm{initial}}^{\mathrm{R\text{-}AD}}$ according to (16) as the first band, denoted by $\mathbf{b}_1^{\mathrm{R\text{-}AD}} = \mathbf{b}_{\mathrm{initial}}^{\mathrm{R\text{-}AD}}$ or (17) as the first band, denoted by $\mathbf{b}_1^{\mathrm{OSP}} = \mathbf{b}_{\mathrm{initial}}^{\mathrm{OSP}}$ and let $\mathbf{\Omega}_1 = \left\{\mathbf{b}_1^{\mathrm{R\text{-}AD/OSP}}\right\}$ be denote either case.

2. For the kth band to be selected with $k \geq 2$, find $\mathbf{b}_k^{\mathrm{R\text{-}AD}}$ by

$$\mathbf{b}_k^{\mathrm{R\text{-}AD/OSP}} = \left\{ \max_{\mathbf{b}_l \in \mathbf{\Omega}_{k-1}^c} \mathbf{b}_l^T \mathbf{R}_{\mathbf{\Omega}_{k-1}^c}^{-1} \mathbf{b}_l \right\} \tag{18}$$

where $\mathbf{\Omega}_{k-1} = \left\{\mathbf{b}_j^{\mathrm{R\text{-}AD/OSP}}\right\}_{j=1}^{k-1}$ and $\mathbf{\Omega}_{k-1}^c = \mathbf{\Omega} - \mathbf{\Omega}_{k-1}$.

3. Check if $k = n_{\mathrm{BS}}$. If yes, the algorithm is terminated. Otherwise, increase k by one, $k \leftarrow k+1$ and go to step 2.

As a final note, a remark is worthwhile. Despite that BP-SBF and BS-SBF cannot be implemented in real-time, they can be implemented in a two-pass process to realize their real-time capability. That is, the first pass is to acquire full bands to implement initial band (IB), BP, and BS. Once all bands are fully processed, it is then followed by a second pass which can be implemented in real time in the same manner that SBF-BSQ is carried out in real time by prioritized bands in the same manner of BSQ.

IV. Discussion on Differences of Band Fusion From BS and PBS

Intuitively, it is natural to think that SBF may be an alternative means of implementing BS and can be considered as PBS. As a matter of fact, SBF, BS, and PBS are three quite different band processing techniques and designed for different rationales on how to take care of interband correlation.

A. Differences From BS

BS has been widely studied to reduce data dimensionality by removing redundant or unwanted bands, while maintaining a certain level of performance. Many various approaches have been developed and reported in the literature. Since the main focus of this section is discussions on its difference from the proposed SBF, a brief review on BS is necessary. Basically, BS can be categorized according to strategies of how to select bands. A simple and also effective approach is BP which ranks all bands according to certain criteria, such as variance, signal-to-noise ratio entropy, information divergence criteria [48], CBS [45], self-mutual information [49], etc. A second approach is band clustering based on grouping bands with similar spectral properties into clusters and then one representative band is selected from each cluster such as [50]. A third approach is to adopt certain band search strategies to find optimal bands in a feed forward or backward manner such as minimum estimated abundance covariance (MEAC) [51], constrained-target BS [52], target-constrained interference-minimized BS [53], etc. A fourth approach is band subset selection (BSS), also referred to as band combination, which selects an optimal band subset of multiple bands as a band combination instead of finding bands one at a time by the third approach. Examples include particle swarm optimization [54], firefly algorithm [55], dominant set extraction BS (DSEBS) [56], multigraph determinantal point process (MDPP) [57], channel capacity-based BSS [58], band subset for anomaly detection [39], constrained multiple BS [59], linearly constrained minimum variance-based BSS [60], and class signature-constrained background (BKG)-suppressed based band selection [61]. Specifically, Wang *et al.* [62] developed an optimal neighborhood reconstruction (ONR)-based BSS approach based on correlated neighborhood property (CNP) to exploit the strong correlation between neighboring bands. It evaluated a band combination by assessing its ability to reconstruct the original data and applied a noise reducer to minimize the influence of noisy bands.

Two major key issues arising in all the above-mentioned approaches are: 1) the requirement of prior knowledge of how many bands needed to be selected, i.e., number of bands to be selected, n_{BS} and 2) how to take care of interband correlation among selected bands. These two issues are very closely related. Despite that BSS is also designed for this purpose, its selected band subsets or combinations are very limited due to the infeasibility of exhausting all possible band combinations as band subsets. As a result, interband correlation cannot be effectively addressed by such a limited and small number of BSS or band combinations. The SBF is particularly developed to address these two issues. In particular, its generated sequential spectral profiles across the range of wavelengths provide detected target abundance fractions band by band which are very valuable information to identify significant bands with no need of BS. In this case, we can select those bands which have abrupt changes in the spectral profile. On the other hand, we can determine n_{BS} when the target detected abundance fractions begin to converge.

1) SBF-BSQ and IBD-SBF are developed to perform band processing without BP.
2) BP-SBF makes BP-based BS a sequential process to provide the interband spectral changes in detected abundance fractions which cannot be offered by BP.
3) BS-SBF makes use of its generated spectral profiles of detected target abundance fractions spectral profiles to perform BS with no need of determining n_{BS}, which is a critical issue for BSS.

It should be noted that all the above four versions of SBF provide the final results of using full bands when they reach the last band.

B. Differences From PBS

As noted in the introduction, SBF-BSQ has its root in PBS which was originally developed for real-time band processing as bands are acquired BSQ. The difference between SBF and PBS is that the former performs BF sequentially band by band as opposed to the latter which performs band processing by adding one band at a time progressively. Specifically, SBF also takes advantage of the recursive forms derived in [2, Chapter 14] to perform BF sequentially. Interestingly, a recent work [63] developed two convolutional neural

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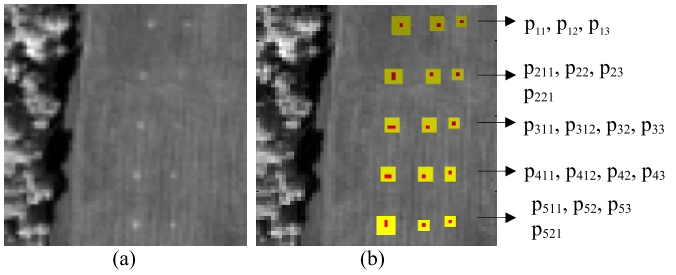


Fig. 1. (a) HYDICE panel scene which contains 15 panels. (b) Ground truth map of spatial locations of 19 R panel pixels.

network (CNN)-based deep learning algorithms for land cover classification using only four bands [RGB + near infrared response (NIR)] and five bands (RGB + NIR + LiDAR) which were augmented using extended multiattribute profiles. This work suggests that SBF can also fuse these two band subsets as an application of SBF where the CNN can be replaced by anomaly detection using SBF to fuse these bands to detect changes in land cover classification. The key issue is how to derive mathematical fusion formulas similar to anomaly detectors for CNN. This is apparently not trivial.

V. Real Image Experiments for SBF

An airborne hyperspectral digital imagery collection experiment (HYDICE) scene shown in Fig. 1 was used to demonstrate the utility of BF.

The data were collected in August 1995 from a flight altitude of 10 000 ft. There are 15 square panels in Fig. 1(a) with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m respectively. Due to the ground sampling distance approximately 1.56 m, the each of panels in the first column except the first row contains two-panel pixels highlighted by red, p_{211}, p_{221} in row 2, p_{311}, p_{312} in row 3, p_{411}, p_{412} in row 4, p_{511}, p_{521} in row 5 as shown in Fig. 1. All the remaining 11 panels in Fig. 1 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, p_{13} in row 1, p_{22}, p_{23} in row 2, p_{32}, p_{33} in row 3, p_{42}, p_{43} in row 4, p_{52}, p_{53} in row 5. Therefore, there are a total of 19 red panel pixels. Fig. 1(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. This data set is particularly used to demonstrate the effectiveness of SBF because SBF produces the sequential band profiles of spectral changes in anomaly maps detected for all 19 R panel pixels regarded as anomalies during the BF process. To the authors' best knowledge, no data set available in the open literature can provide such detailed ground truth information that this HYDICE data set can. By taking advantage of this data set, the following experiments were designed and conducted:

1) performing SBF-BSQ starting with the first band;
2) performing FCIBD-SBF which implements IBD-SBF in a forward and circular fashion where the initial band were selected by (16) and (17);
3) performing R-BP-SBF using (13) and OSP-BP-SBF using (15);
4) performing R-BS-SBF and OSP-BS-SBF.

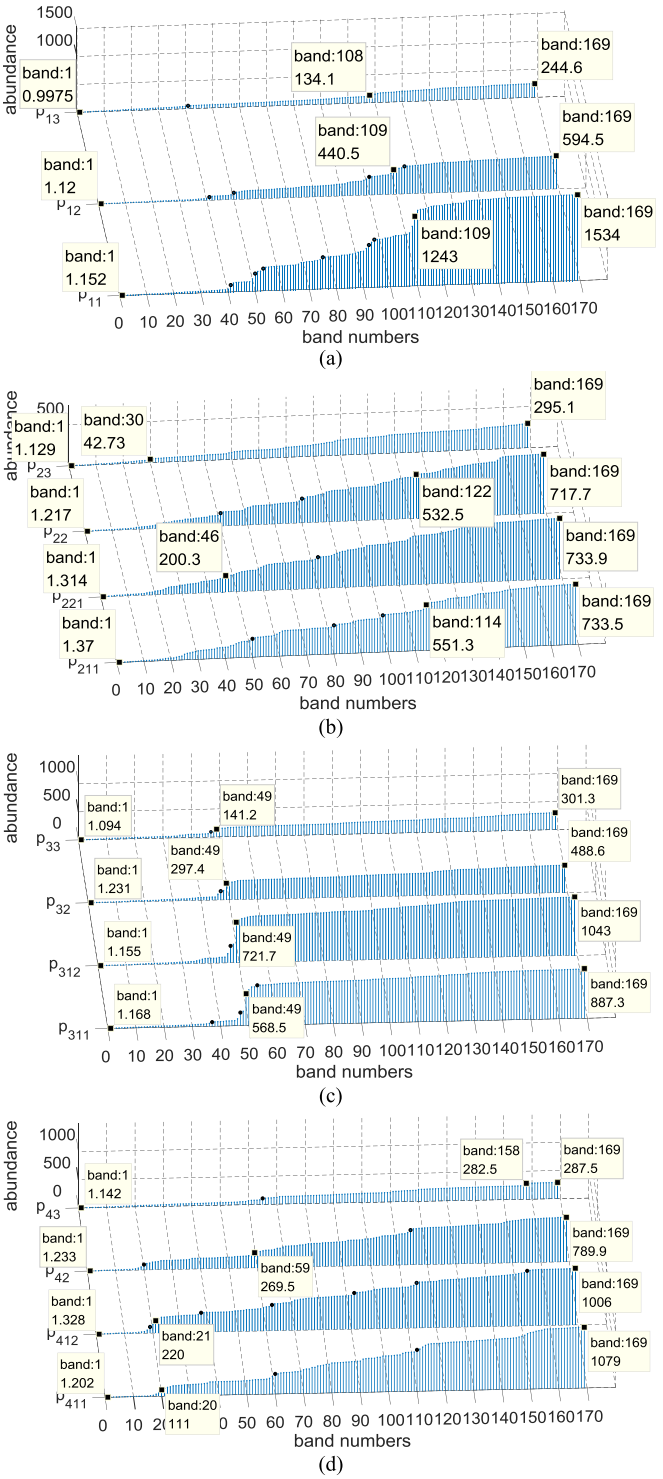


Fig. 2. Sequential abundance fraction plots of 19 R panel pixels detected by R-AD using SBF-BSQ. (a) Abundance fraction plot of panel pixels in row 1. (b) Abundance fraction plot of panel pixels in row 2. (c) Abundance fraction plot of panel pixels in row 3. (d) Abundance fraction plot of panel pixels in row 4. (e) Abundance fraction plot of panel pixels in row 5.

A. Sequential SBF According to BSQ

The first experiment performed SBF according to BSQ (SBF-BSQ) starting with the first band. The 19 R panel pixels of HYDICE were detected by the R-AD detector in (2) as anomalous pixels. According to BSQ, the data were processed sequentially band-by-band from the first band as the initial

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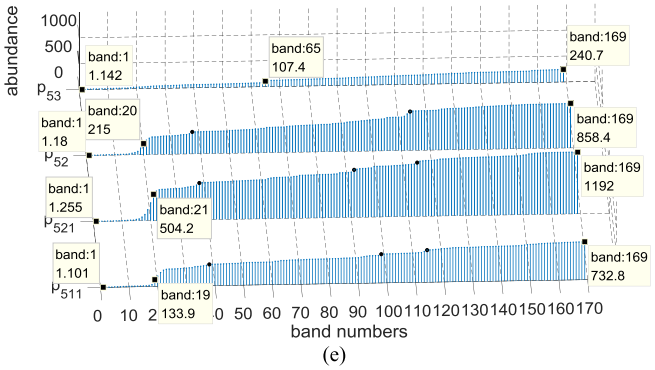


Fig. 2. *(Continued.)* Sequential abundance fraction plots of 19 R panel pixels detected by R-AD using SBF-BSQ. (a) Abundance fraction plot of panel pixels in row 1. (b) Abundance fraction plot of panel pixels in row 2. (c) Abundance fraction plot of panel pixels in row 3. (d) Abundance fraction plot of panel pixels in row 4. (e) Abundance fraction plot of panel pixels in row 5.

band, while the BF was performed at the same time. The first band was immediately used for anomaly detection by R-AD right after it was acquired. The data of the second band were then fused by (7) with the anomaly detection map obtained by the first band at the same time, while the second band was collected. The same procedure was continued to perform band acquisition and BF by (12) at the same time until all bands were acquired and fused to produce sequential band profiles of R-AD detected abundance fractions of 19 R panel pixels by SBF-BSQ as shown in Fig. 2. It should be noted that the detected abundance fractions of 19 R panel pixels were first calculated by fusing the initial bands while their second bands being acquired via (7). This yielded two-band subsets to be fused with their third bands. In this case, the detected abundance fractions of 19 R panel pixels were further calculated by (12) to fuse the resulting two-band subsets with their third bands and then the same process was continued on by (12) until it reached the last band. The bands highlighted in Fig. 2 by small dark squares indicated sudden jump peaks of detected abundance fractions with their corresponding values. From Fig. 2, the detected abundance fractions of all 19 R panel pixels were monotonically increasing as more bands were fused. When a band occurred with its detected abundance fraction suddenly jumped as a peak and then the detected abundance fractions of subsequent bands right after the band remained nearly the same and unchanged. In this case, this specific band would be considered as a significant band. Such phenomenon was particularly evidential in the detection of R panel pixels in rows 3, 4, and 5.

It is also worth noting that the anomaly detection results by R-AD via SBF-BSQ were identical to that obtained by using all bands to be fused. Fig. 3(a) shows the final anomaly map detected by R-AD using SBF-BSQ along with Fig. 3(b) which shows its binary map of Fig. 3(a) obtained by Otsu's method.

Several benefits can be gained from using SBF-BSQ. First of all, it provides sequential band profiles of R-AD detection maps as shown in Fig. 2 where significant bands for anomaly detection can be identified by the sudden jump peaks of their detected abundance fractions. Second, R-AD can be carried out band-by-band sequentially without waiting for all bands being acquired. Third, it offers an alternative approach to

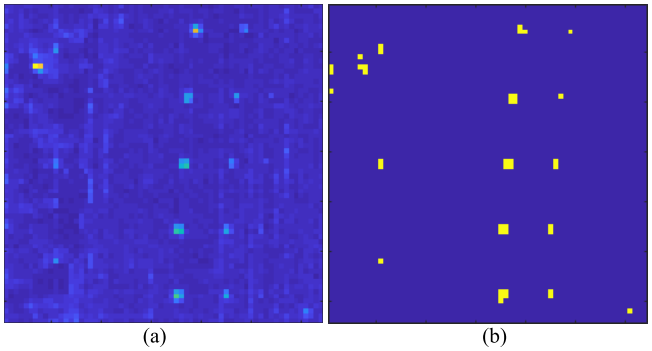


Fig. 3. Detection maps of R-AD using SBF-BSQ as along with its binary map obtained by Otsu's method. (a) R-AD using SBF-BSQ. (b) Binary map of Fig. 3(a) by Otsu's method.

BS where SBF-BSQ does not need to know the number of bands to be selected, n_{BS}, while BS does. Particularly, with no constrained by n_{BS}, SBF-BSQ can extract crucial bands across the entire spectral wavelengths for anomaly detection.

B. IBD-SBF

A second experiment was designed to demonstrate the impact of the initial BS on anomaly detection performance when IBD-SBF is implemented. It started with a particularly selected band as its initial band and then progresses SBF. The advantage of IBD-SBF over SBF-BSQ is that it could quickly reach the same performance as SBF-BSQ with fewer fused bands. However, IBD-SBF is also traded for the advantage of BSQ which allows SBF to process bands in real-time band-by-band without waiting for full bands being acquired. This is because IBD-SBF requires full band information to select a best possible band as its initial band to begin with. Fig. 4(a) and (b) provides sequential band profiles of R-AD detected abundance fractions of 19 R panel pixels by FCIBD-SBF using $\mathbf{B}_{\mathrm{inital}}$ selected by (16) and (17) as band 53 and band 125, respectively. Apparently, a different initial band produced a different band profile of detected abundance fractions of the 19 R panel pixels. Compared to the sequential band profiles produced by SBF-BSQ in Fig. 2, the bands which detected high abundance fractions of 19 R panel pixels generally occurred much earlier with fusing fewer bands than their counterparts in Fig. 2.

Like SBF-BSQ the sequential abundance fractions of 19 R panel pixels detected in Fig. 4 were always monotonically increasing. However, unlike SBF-BSQ which used the first band as the initial band, IBD-SBF detected the abundance fractions of nearly all the 19 R panel pixels significantly increased right after fusing the initial band. Most importantly, using a particularly selected initial band, FCIBD-SBF could fuse fewer bands to obtain higher anomaly detection abundance fractions. For example, as shown in Fig. 4(a1) and (b1), the abundance fractions in the first row have been greatly increased after fusing about 60 bands and 30 bands using R-BP and OSP-BP, respectively. By contrast, SBF-BSQ only began to increase the detected abundance fractions of panel pixels in the first row after fusing 40 bands. Table I tabulates the abundance fractions of panel pixels using one band in the first detection of SBF-BSQ compared to IBD-SBF using R-BP and OSP-BP.

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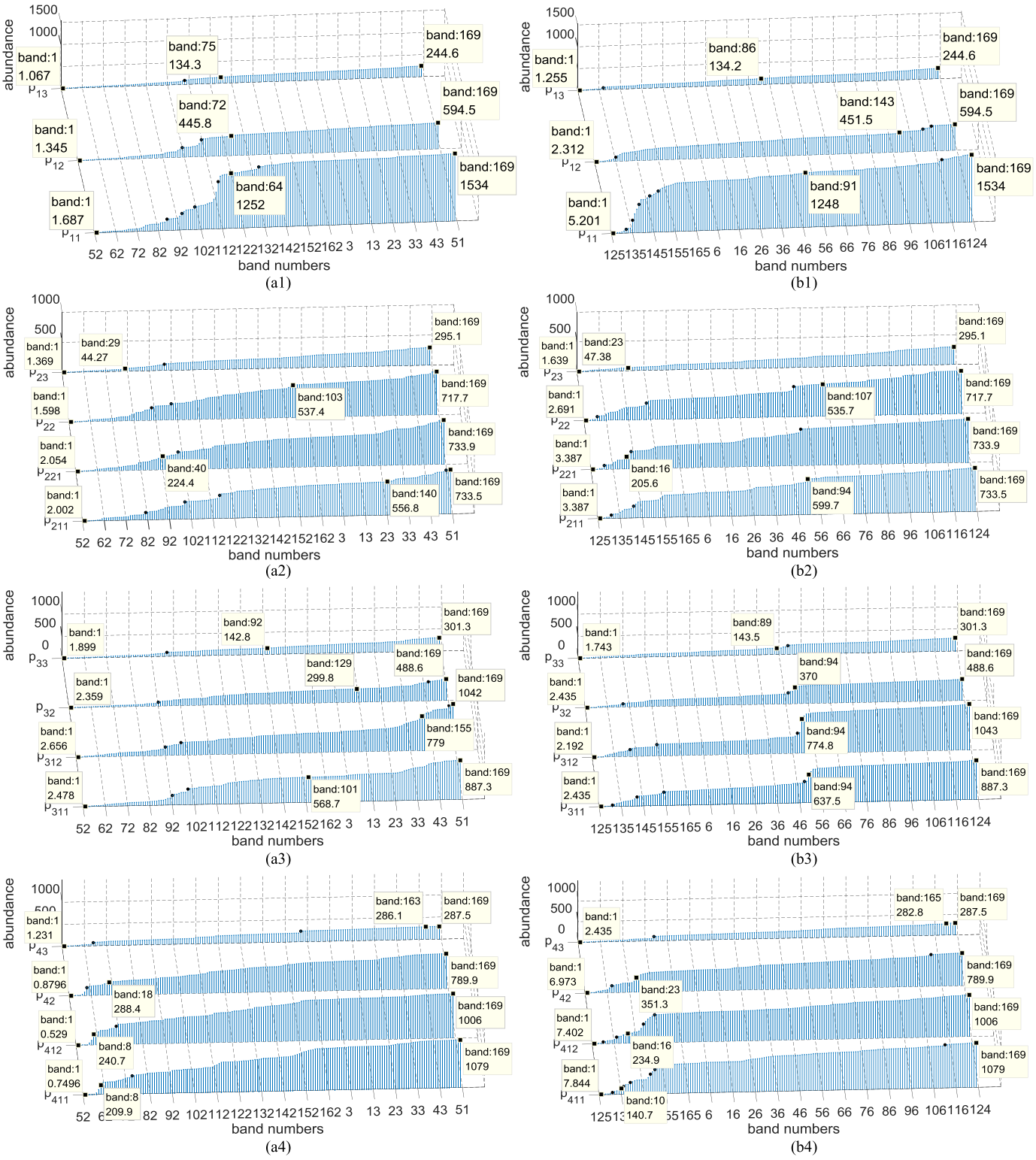


Fig. 4. Spectral variations of anomaly detection abundance of 19 R panel pixels of HYDICE in the process of IBD-SBF based on R-BP and OSP-BP. (a1)–(a5) FCIBD-SBF based on R-BP. (b1)–(b5) FCIBD-SBF based on OSP-BP.

As can be seen from Table I, the band numbers in bold are the highest abundance detected by BP-SBF, R-FCIBD-SBF, and OSP-IBD-SBF. In general, IBD-SBF produced higher anomaly detection abundances than SBF-BSQ. For example, OSP-BP IBD-SBF produced higher abundance values when detecting the panel pixels in the fourth and fifth rows, both of which were about seven times that of SBF-BSQ, specifically, the abundance of the panel pixel p_{521} which reached 11.84. The higher the abundance is the anomaly as panel pixel, the easier is the panel pixel to be detected. Because the

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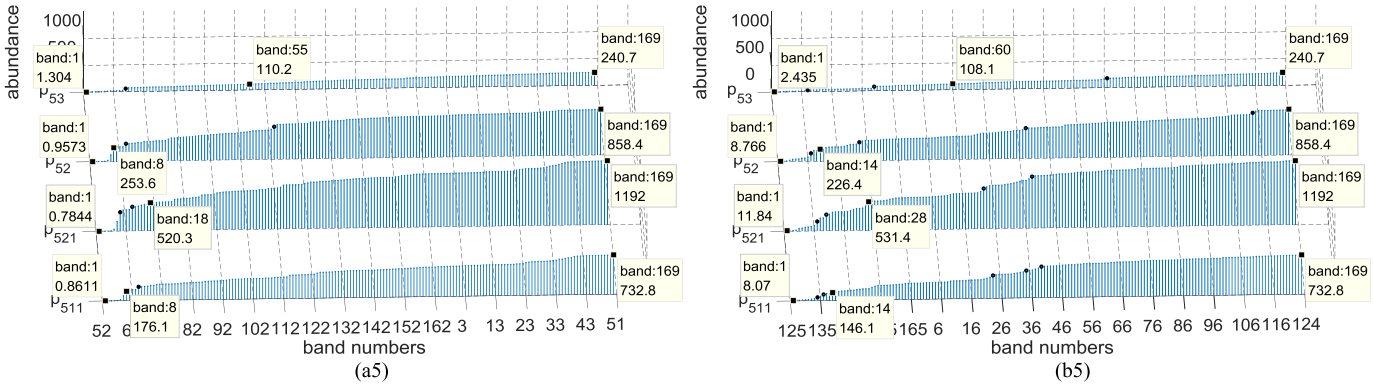


Fig. 4. *(Continued.)* Spectral variations of anomaly detection abundance of 19 R panel pixels of HYDICE in the process of IBD-SBF based on R-BP and OSP-BP. (a1)–(a5) FCIBD-SBF based on R-BP. (b1)–(b5) FCIBD-SBF based on OSP-BP.

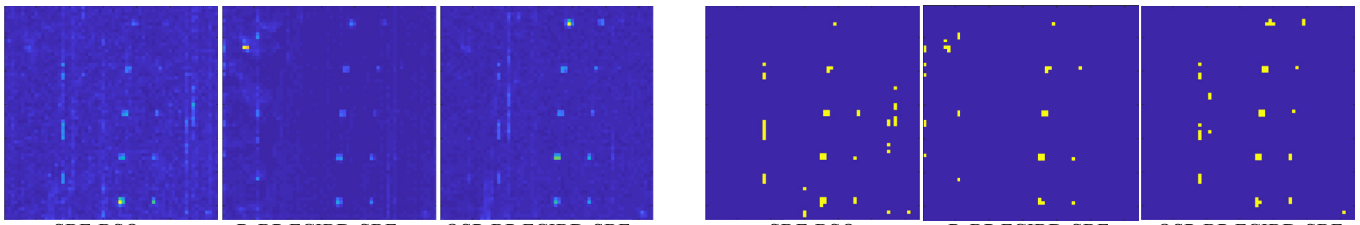


Fig. 5. Results and binarized image of SBF-BSQ, IBD-SBF based on R-BP, and IBD-SBF based on OSP-BP using 47 bands. (a) R-AD detection maps. (b) Binary maps thresholded by Otsu's method.

TABLE I
ABUNDANCE FRACTIONS OF 19 R PANEL PIXELS USING FCIBD-SBF

No.	Abundance fractions of 19 R panel pixels		
	SBF-BSQ (B_{inital} = 1)	R-BP FCIBD-SBF (B_{inital} = 53)	OSP-BP FCIBD-SBF (B_{inital} = 125)
p_{11}	1.152	1.687	**5.201**
p_{12}	1.12	1.345	**2.312**
p_{13}	0.9975	1.067	**1.255**
p_{211}	1.37	2.002	**3.387**
p_{221}	1.314	2.054	**3.387**
p_{22}	1.217	1.598	**2.691**
p_{23}	1.129	1.369	**1.639**
p_{311}	1.168	**2.478**	2.435
p_{312}	1.155	**2.656**	2.192
p_{32}	1.231	2.359	**2.435**
p_{33}	1.094	**1.899**	1.743
p_{411}	1.202	0.7496	**7.844**
p_{412}	1.328	0.529	**7.402**
p_{42}	1.233	0.8796	**6.973**
p_{43}	1.142	1.231	**2.435**
p_{511}	1.101	0.8611	**8.07**
p_{521}	1.255	0.7844	**11.84**
p_{52}	1.18	0．9573	**8.766**
p_{53}	1.142	1.304	**2.435**

TABLE II
NUMBER OF FUSED BANDS WHEN REACHING THE SAME ABUNDANCE

No.	Number of used bands		
	SBF-BSQ (B_{inital} = 1)	R-BP FCIBD-SBF (B_{inital} = 53)	OSP-BP FCIBD-SBF (B_{inital} = 125)
p_{11}	109	**64**	91
p_{12}	109	**72**	143
p_{13}	108	**75**	86
p_{211}	114	140	**94**
p_{221}	46	40	**16**
p_{22}	122	**103**	107
p_{23}	30	29	**23**
p_{311}	**49**	101	94
p_{312}	**49**	155	94
p_{32}	**49**	129	94
p_{33}	**49**	92	89
p_{411}	20	**8**	10
p_{412}	21	**8**	16
p_{42}	59	**18**	23
p_{43}	**158**	163	165
p_{511}	19	**8**	14
p_{521}	21	**18**	28
p_{52}	20	**8**	14
p_{53}	65	**55**	60

abundance changes monotonically increasing, the abundance detected by the initial band will have a direct impact on the trend of subsequent BF, thus, will affect the panel detection. That is, the higher is the abundance fraction detected by the initial band, the fewer fused bands are to be fused to achieve desired anomaly detection performance.

Table II shows an example which lists the number of fused bands of SBF-BSQ, FCIBD-SBF using R-BP and OSP-BP when the abundance reached the same abundance fraction.

From Table II, the number of bands required by the three IBD-SBF methods, SBF-BSQ, R-BP FCIBD-SBF, and OSP-BP FCIBD-SBF to achieve the same abundance was different where the boldfaced number is the minimum number of bands required by the three methods when the detected abundance fractions reached approximate same values. In general,

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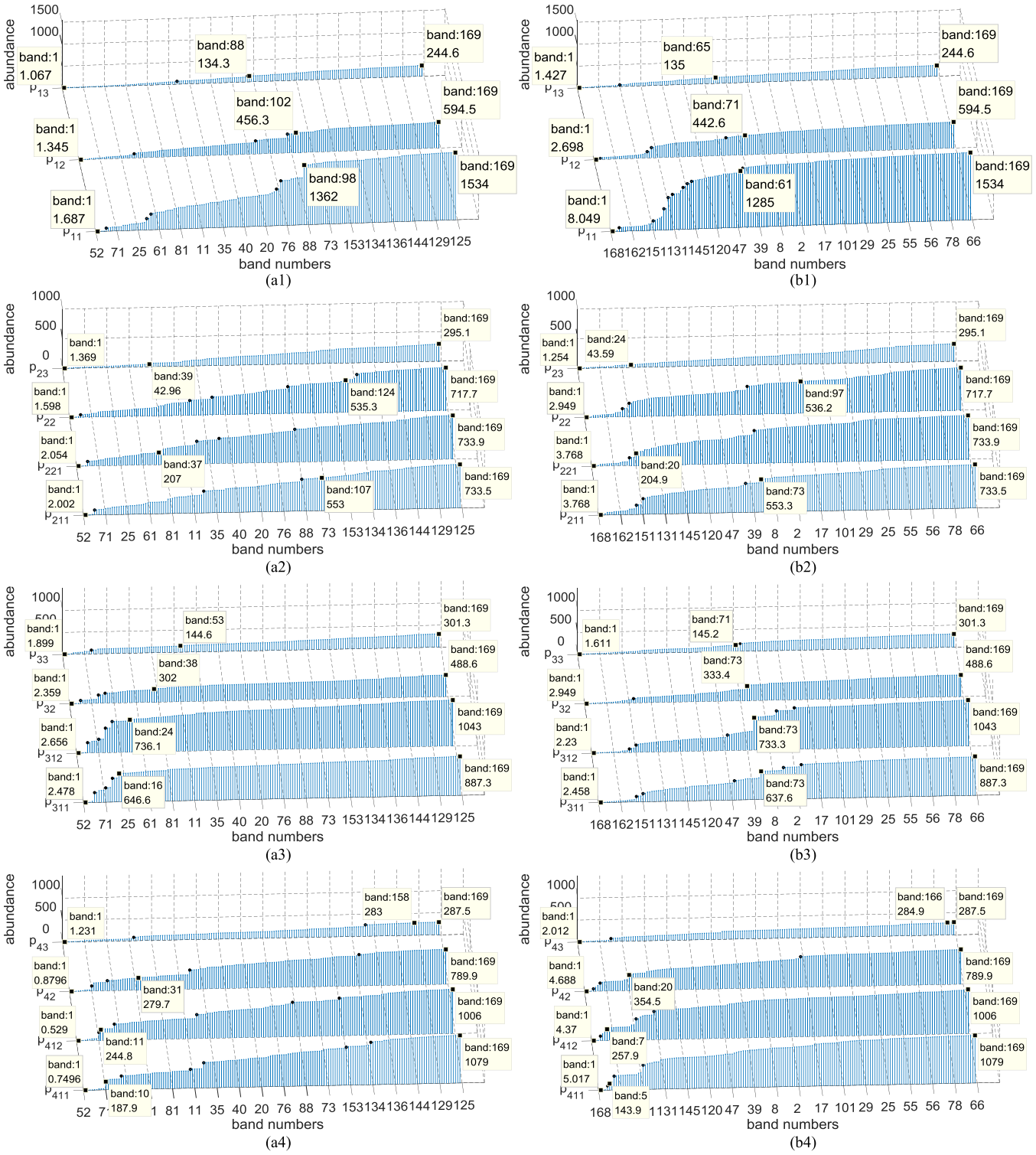


Fig. 6. Sequential detected abundance fractions of 19 R panel pixels produced by R-BP-SBF and OSP-BP-SBF. (a1)–(a5) R-BP-SBF. (b1)–(b5) OSP-BP-SBF.

IBD-SBF used fewer fused bands to achieve the approximate abundance fractions by SBF-BSQ. Fig. 5 shows the detection results of the three methods using the same number of bands, 47 bands where the image was binarized by the Otsu method and most of the anomalies can be detected by three methods where OSP-BP FCIBD-SBF could detect all the R panel pixels in the first two columns using only 47 bands with very effective BKG suppression.

C. BP-SBF

A third experiment performed BP-SBF using (13) and (15) as BP criteria where all the bands were ranked by BP prior

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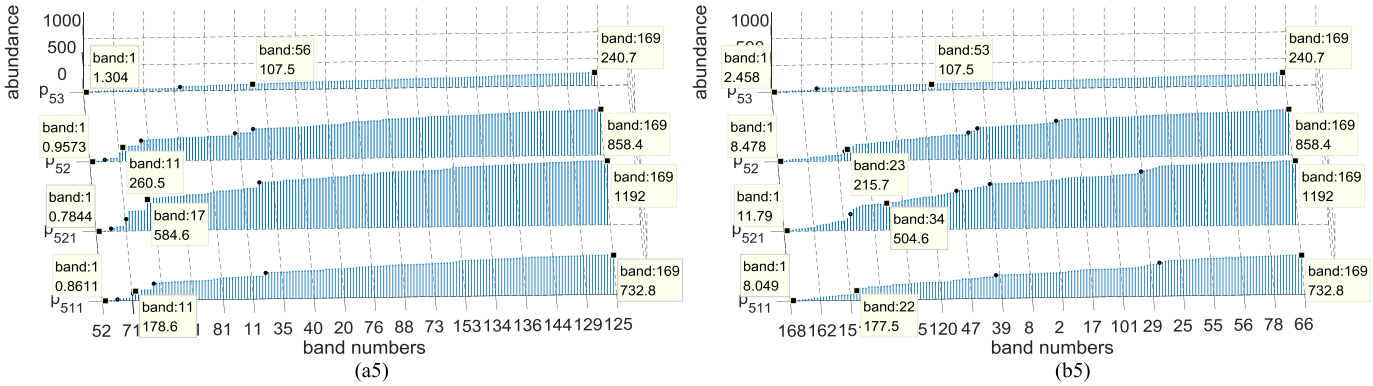


Fig. 6. *(Continued.)* Sequential detected abundance fractions of 19 R panel pixels produced by R-BP-SBF and OSP-BP-SBF. (a1)–(a5) R-BP-SBF. (b1)–(b5) OSP-BP-SBF.

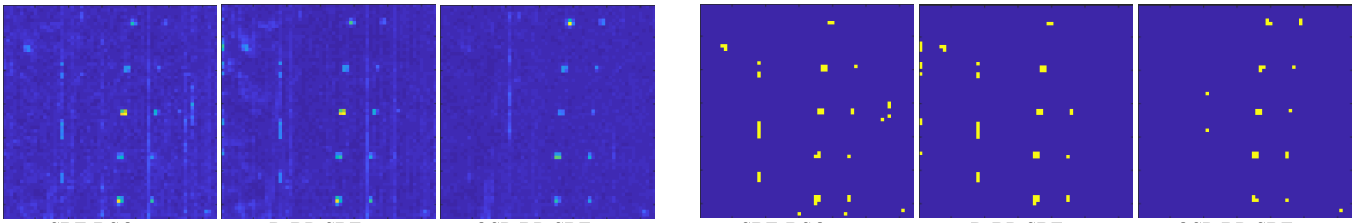


Fig. 7. Results and binarized images of SBF-BSQ, R-BP-SBF, and OSP-BP-SBF using the corresponding number bands. (a) R-AD detection maps. (b) Binary maps thresholded by Otsu's method.

to BF. Fig. 6 shows sequential detected abundance fractions of 19 R panel pixels produced by R-BP-SBF and OSP-BP-SBF. Table III also lists the number of bands required by SBF-BSQ, R-BP-SBF, and OSP-BP-SBF as they reached approximately the nearly same abundance fractions. Analogous to SBF-BSQ and IBD-SBF, the R-AD detected abundance fractions produced by BP-SBF were also monotonically increasing as more bands were fused. It is worth noting that in the experiment of IBD-SBF, the change of the initial band did not have a sufficient benefit from the fusion process in detecting the panel pixels in the third row. This problem has been improved in this experiment. As shown in Fig. 6(a3), R-BP-SBF obtained a higher abundance value right after fusing only 16 bands, and the abundance value was remained in a steady increase after subsequent bands were fused. The average number of bands used by each method was 59, 42, and 44, respectively.

Using the bands in Table III as the stopping bands for SBF, Fig. 7 shows the detection results of SBF-BSQ, R-BP-SBF, and OSP-BP-SBF with their binary maps thresholded by Otsu's method where most of panel pixels were detected.

Comparing the results in Fig. 7 to that in Fig. 5 produced by SBF-BSQ and IBD-SBF, R-BP-SBF and OSP-BP-SBF could detect the same panel pixels using the fewer fused band and also had better BKG suppression. This advantage was well reflected in the OSP-BP-SBF detection results where most of panel pixels were detected, and only very few pixels in the BKG that were miss-detected as panel pixels by OSP-BP-SBF.

From the above results, the abundance fractions of panel pixels as anomalies reached high values after the first half of the bands were fused, and then remained relatively stable in the last half. This indicated that it was not necessary to fuse all the bands to detect panel pixels as anomalies. This observation leads to the following experiment.

TABLE III
NUMBER OF BANDS USED TO REACH APPROXIMATE THE SAME ABUNDANCE FRACTIONS

No.	Number of used bands		
	SBF-BSQ	R-BP-SBF	OSP-BP-SBF
p_{11}	110	98	**62**
p_{12}	108	98	**62**
p_{13}	108	88	**65**
p_{211}	112	94	**66**
p_{221}	97	84	**72**
p_{22}	107	98	**73**
p_{23}	92	**81**	87
p_{311}	49	**16**	73
p_{312}	48	**13**	61
p_{32}	48	**13**	50
p_{33}	47	**16**	56
p_{411}	21	10	**5**
p_{412}	21	11	**7**
p_{42}	21	10	**6**
p_{43}	60	32	**16**
p_{511}	19	**11**	22
p_{521}	20	**11**	23
p_{52}	20	**11**	23
p_{53}	21	19	**15**

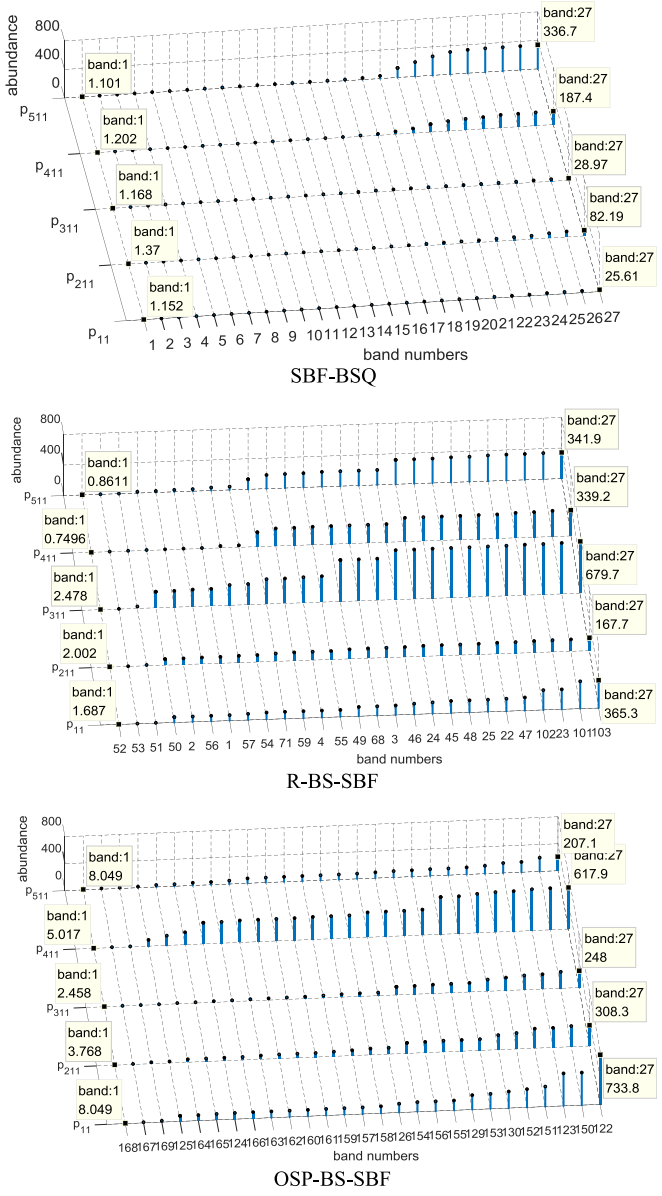
D. BS-SBF

A fourth experiment performed BS-SBF using R-BS-SBF and OSP-BS SBF. Recall that BS-SBF differs from BP-SBF by two key features. One is that it does not require

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TABLE IV
27 BANDS OF SBF-BSQ, R-BS-SBF, AND OSP-BS-SBF

SBF-BSQ	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
R-BS-SBF	52	53	51	50	2	56	1	57	54	71	59	4	55	49	68	3	46	24	45	48	25	22	47	102	23	101	103
OSP-BS-SBF	168	167	169	125	164	165	124	166	163	162	160	161	159	157	158	126	154	156	155	129	153	130	152	151	123	150	122



SBF-BSQ

R-BS-SBF

OSP-BS-SBF

Fig. 8. Detected abundance fractions of five-panel pixels in the first column of HYDICE data set fused by SBF-BSQ, R-BS-SBF, and OSP-BS-SBF.

prioritizing all the bands as BP-SBF does. But it does need to know the number of bands to be selected, n_{BS}, which is not required by BP-SBF. For this purpose, virtual dimensionality (VD) developed in [2, Chapter 4] and [64]–[67] was used to determine n_{BS} which was 9 and 18. However, in order to see better performance of R-BS-SBF and OSP-BS-SBF, we extended n_{BS} to 27 for further analysis. Table IV lists the first 27 bands selected by SBF-BSQ, R-BS-SBF, and OSP-BS-SBF.

As an illustrative example, Fig. 8 only shows the detected abundance fractions of the first five-panel pixels in the first column of HYDICE data produced by the three BS methods. As can be seen from Fig. 8, BS-SBF could fuse fewer bands to obtain higher abundance fractions than SBF-BSQ did. There was little change in the abundance fractions detected by SBF-BSQ fusing the first 18 bands. By contrast, BS-SBF selected bands according to R-BS and OSP-BS, the detected abundance started to climb right after fusing the first four selected bands. Second, since the number of bands selected has been determined in advance, BS-SBF did not need to fuse all the bands, and only needed to fuse the desired n_{BS} bands ranked by BS.

TABLE V
ABUNDANCE FRACTIONS OF PANEL PIXELS IN THE FIRST COLUMN BY R-AD USING SBF-BSQ, R-BS-SBF, AND OSP-BS-SBF WHEN $n_{\mathrm{BS}} = 9, 18, 27$

n_{BS}	No.	detected abundance fractions		
		SBF-BSQ	SBF-RADBS	SBF-OSPBS
9	p_{11}	5.452	89.24	**92.84**
	p_{211}	8.452	**90.98**	55.4
	p_{311}	9.757	**270**	24.93
	p_{411}	3.577	25.76	**325.7**
	p_{511}	10.27	42.83	**50.68**
18	p_{11}	11.17	**148.2**	143.8
	p_{211}	25.08	138.2	**158.9**
	p_{311}	22.82	**647.2**	123
	p_{411}	38.73	311.4	**370.6**
	p_{511}	37.74	**318.7**	92.94
27	p_{11}	25.61	365.3	**733.8**
	p_{211}	82.19	167.7	**308.3**
	p_{311}	28.97	**679.7**	248
	p_{411}	187.4	339.2	**617.9**
	p_{511}	336.7	**341.9**	207.1

Table V lists the abundance fractions detected by R-AD using SBF-BSQ, R-BS-SBF, and OSP-BS-SBF when n_{BS} was set to 9, 18, 27.

Fig. 9 shows the fused detection results along with their corresponding Otsu's method thresholded binary maps when $n_{\mathrm{BS}} = 9, 18, 27$. From Fig. 9, when $n_{\mathrm{BS}} = 9$, most of the panel pixels were not detected by the three methods due to a small number of bands to be fused, and the BKG suppression was poor. Only OSP-BS-SBF detected some panel pixels in the fourth and fifth rows. As n_{BS} was increased to 18, more panel pixels were detected, but the BKG pixels that were also falsely detected as anomalies. When the n_{BS} was further increased to 27, the BKG was significantly suppressed with very few falsely alarmed pixels. In particular, OSP-BS-SBF not only achieved the same performance as that using full bands did, but also had better BKG suppression.

E. Computational Complexity

The computational time varies with different methods. Since the matrix inversion, i.e., $\mathbf{R}^{-1}$ is involved in the anomaly

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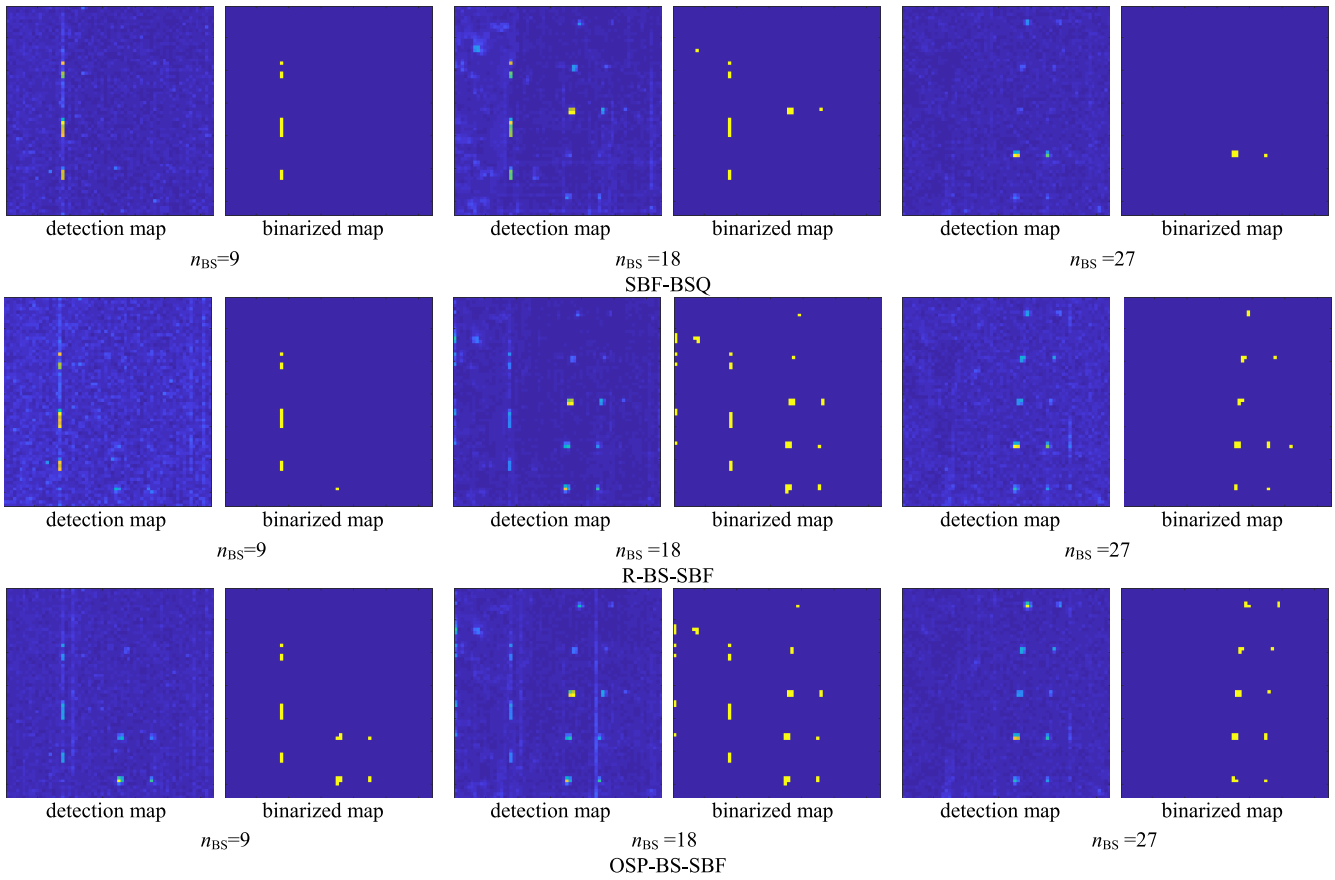


Fig. 9. Fused detection results and binarized image of SBF-BSQ, R-BS-SBF, and OSP-BS-SBF when $n_{BS} = 9, 18, 27$.

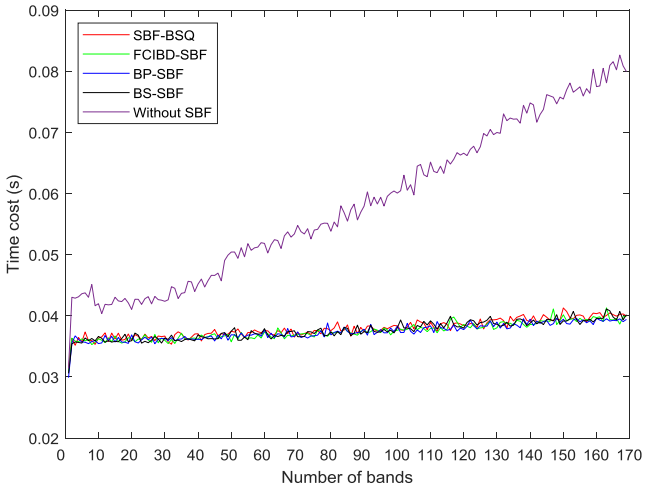


Fig. 10. Comparison of computational times among BS and various SBF methods.

detection process, the computational time is determined by the number of bands used in $\mathbf{R}^{-1}$. The more the bands used, the more complex the calculation of $\mathbf{R}^{-1}$ and thus, the longer the computation time for $\mathbf{R}^{-1}$. The proposed SBF perfectly solves this problem where the anomaly detection is performed band by band without matrix inversion instead of all selected bands being used in calculating $\mathbf{R}^{-1}$ at the same time. Consequently, the computational time of SBF in anomaly detection is not affected by the number of bands as more bands are being processed sequentially. Fig. 10 shows a comparative analysis of computational time among BS and four versions of SBF as the number of bands is increased where all experiments were run on a computer with Intel Core i7-6700, 8-GB memory, the OS is Windows 10, and the platform is MATLAB R2018a.

As noted in Fig. 10, the computational time required by BS is increased exponentially. By contrast, the four versions of SBF which required nearly the same computational time which is almost flat as a constant when the number of bands grows. This indicated that when it comes to SBF, the four versions of SBF made no appreciable differences in computational complexity.

VI. Conclusion

This article develops a new approach to hyperspectral BF, referred to as SBF which is derived from the data acquisition BSQ format. Four versions of SBF, SBF-BSQ, IBD-SBF, BP-SBF, and BS-SBF are particularly designed to realize SBF. In order to validate the fused results by SBF identical to the results using all bands to be fused, the mathematical theory is also derived and analyzed for SBF. The benefits and advantages of using SBF over BS are also demonstrated by extensive experiments via anomaly detection. Several contributions of this article are considered to be significant and summarized as follows.

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1) First and foremost, SBF is quite different from data fusion where the SBF process takes place in the anomaly detector used to fuse different band subsets as opposed to data fusion which fuses data sets does. The derivations in Appendixes are new.
2) Despite the fact that SBF-BSQ was originally derived from PBP/PBS, the other three versions, IBD-SBF-BSQ, BP-SBF, and BS-SBF, developed for SBF are new and have never explored in the past.
3) Two BP criteria, correlation matrix-based and OSP-based criteria, are developed to prioritize bands used by SBF, both of which are new and have not been proposed in the literature.

APPENDIX A

$$\begin{aligned}
&\delta^{\text{R-AD}}_{\mathbf{b}_s\mathbf{b}_t} \\
&= \mathbf{r}^T_{\mathbf{b}_s\mathbf{b}_t} R^{-1}_{\mathbf{b}_s\mathbf{b}_t} \mathbf{r}_{\mathbf{b}_s\mathbf{b}_t} = \left(\mathbf{r}^T_{\mathbf{b}_s}, \mathbf{r}_{\mathbf{b}_t}\right) \\
&\quad \times \begin{bmatrix} R^{-1}_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} & -\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \\ -\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} & N\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \end{bmatrix} \begin{pmatrix} \mathbf{r}_{\mathbf{b}_s} \\ \mathbf{r}_{\mathbf{b}_t} \end{pmatrix} \\
&= \mathbf{r}^T_{\mathbf{b}_s} R^{-1}_{\mathbf{b}_s} \mathbf{r}_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}^T_{\mathbf{b}_s} \upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}_{\mathbf{b}_s} \\
&\quad - \mathbf{r}_{\mathbf{b}_t} \beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}_{\mathbf{b}_s} - \beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}^T_{\mathbf{b}_s} \upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}_{\mathbf{b}_t} \\
&\quad + N\mathbf{r}_{\mathbf{b}_t} \beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}_{\mathbf{b}_t} \\
&= \delta^{\text{R-AD}}_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \left(\mathbf{r}^T_{\mathbf{b}_s} \upsilon_{\mathbf{b}_s\mathbf{b}_t\mathbf{b}_s} - N\mathbf{r}_{\mathbf{b}_t}\right) \\
&\quad \times \left(\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \mathbf{r}_{\mathbf{b}_s} - N\mathbf{r}_{\mathbf{b}_t}\right) \qquad \text{(A1)}
\end{aligned}$$

where $\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} = R^{-1}_{\mathbf{b}_s}\mathbf{x}^T_{\mathbf{b}_s}\mathbf{x}_{\mathbf{b}_t}$, and $\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} = (\mathbf{x}^T_{\mathbf{b}_t}[P^{\perp}_{\mathbf{X}^T_{\mathbf{b}_s}}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

APPENDIX B

$$\begin{aligned}
&\delta^{\text{R-AD}}_{\mathbf{B}_{s,m}\mathbf{b}_t} \\
&= \mathbf{r}^T_{\mathbf{B}_{s,m}\mathbf{b}_t} \mathbf{R}^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t} \mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t} = \left(\mathbf{r}^T_{\mathbf{B}_{s,m}}, \mathbf{r}_{\mathbf{b}_t}\right) \\
&\quad \times \begin{bmatrix} \mathbf{R}^{-1}_{\mathbf{B}_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} & -\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \\ -\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} & N\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \end{bmatrix} \\
&\quad \times \begin{pmatrix} \mathbf{r}_{\mathbf{B}_{s,m}} \\ \mathbf{r}_{\mathbf{b}_t} \end{pmatrix} \\
&= \mathbf{r}^T_{\mathbf{B}_{s,m}} \mathbf{R}^{-1}_{\mathbf{B}_{s,m}} \mathbf{r}_{\mathbf{B}_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}^T_{\mathbf{B}_{s,m}} \upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}_{\mathbf{B}_{s,m}} \\
&\quad - \mathbf{r}_{\mathbf{b}_t} \beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}_{\mathbf{B}_{s,m}} - \beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}^T_{\mathbf{B}_{s,m}} \upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}_{\mathbf{b}_t} \\
&\quad + N\mathbf{r}_{\mathbf{b}_t} \beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}_{\mathbf{b}_t} \\
&= \delta^{\text{R-AD}}_{B_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \left(\mathbf{r}^T_{\mathbf{B}_{s,m}} \upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} - N\mathbf{r}_{\mathbf{b}_t}\right) \\
&\quad \times \left(\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \mathbf{r}_{\mathbf{B}_{s,m}} - N\mathbf{r}_{\mathbf{b}_t}\right) \qquad \text{(B1)}
\end{aligned}$$

where $\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} = \mathbf{R}^{-1}_{\mathbf{B}_{s,m}}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{x}_{\mathbf{b}_t}$, and $\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} = (\mathbf{x}^T_{\mathbf{b}_t}[P^{\perp}_{\mathbf{X}^T_{\mathbf{B}_{s,m}}}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

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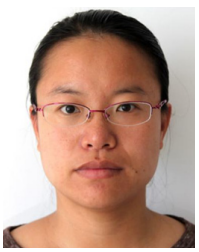
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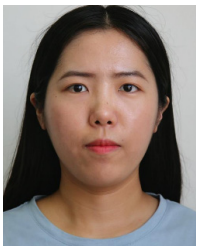
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